MIL-C-24643/46B
3 October 1986
SUPERSEDING
MIL-C-0024643/46A(SH)
8 November 1985
MIL-C-24643/46
28 September 1984

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 600 VOLTS, TYPE LS2WAU (INCLUDING VARIATION LS2WA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the cable described herein shall consist of this specification and the latest issue of MIL-C-24643.

This cable can be manufactured in two variations: LS2WAU (unarmored) and LS2WA (armored).

REQUIREMENTS:

Qualification required.

Construction (watertight)

First - Copper conductor, tin coated, in accordance with ASTM B 286, size 22-7. Maximum diameter 0.033 inch.
Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third - Cross-linked polyethylene insulation. Nominal thickness 0.024 inch. Telephone identification code applied by method 3.
Fourth - Two conductors cabled together to form a pair, with nominal lay of 1-1/2 inches.
Fifth - Forty pairs cabled together, with a lay not greater than 16 times the overall diameter of the layer. Cabling sequence shall be consecutive, starting with pair no. 1, from center outward. Fillers shall be employed to form a firm, well-rounded assembly.
Sixth - Binder tape, applied helically with overlap, at manufacturer's option.
Seventh - Braided shield of AWG no. 34 tin coated copper. Minimum coverage of 85 percent.
Eighth - Binder tape applied helically with overlap.

(B) denotes changes.

AMSC N/A FSC 6145
DISTRIBUTION STATEMENT A Approved for public release; distribution unlimited

Source: https://assist.dla.mil - Downloaded: 2026-10-20T00:13Z
Check the source to verify that this is the current version before use.

Construction variant for type LS2WAU

Ninth - Cross-linked polyolefin jacket. Nominal thickness 0.060 inch. Cable surface marking required.

Construction variant for type LS2WA

Ninth - Cross-linked polyolefin jacket. Nominal thickness 0.060 inch.
Tenth - Braided metal armor and paint.

TABLE I. Details.

Military 1/ part no. M24643/46	Size 2/	Number of pairs	Overall diameter (inches)			
			Type LS2WAU		Type LS2WA	
			Minimum	Maximum	Minimum	Maximum
-01U0	40	40	1.320	1.370	1.370	1.420

1/ Unarmored, overall shield. When armor option is required, see MIL-C-24643 for configuration of part number.
2/ Types LS2WAU and LS2WA.

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet, at 25°C, maximum.	17.71
Voltage withstand - volts, root mean square, minimum	
Conductor to shield	2500
Shield to water	1000
Insulation resistance - megohms/1000 feet, minimum	
Conductor to conductor	5000
Jacket flaws	No failure
Conductor and shield continuity	No failure
Group A:	
Visual and dimensional	No failure
Capacitance - at 1 MHz, pF/ft maximum	
Mutual capacitance	20
Watertightness - see MIL-C-24643 for limits of water leakage	No failure

Source: https://assist.dla.mil -- Downloaded: 202610-20T00:13Z
Check the source to verify that this is the current version before use.

EXAMINATION AND TESTS: (Continued)

	Requirements
Group B:	
Cold bending, cable - at 0 ± 2°C, mandrel maximum diameter 12 times cable diameter	No damage
Drip - 95 ± 1°C	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Physicals (unaged)	
Insulation	
Tensile strength - lb/in^2, minimum	1800
Elongation - percent, minimum	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Group C:	
Physicals (aged)	
Insulation (conductor)	
Air oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged, minimum	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged, minimum	60
Hot oil immersion	
Tensile strength - lb/in^2, minimum	50
Elongation - percent, minimum	50
Shrinkage	No failure
Permanence of printing (jacket) type LS2WAU - cycles, minimum	250
Heat distortion - percent of unaged, maximum	30
Cable filler removability	No failure
Armor - conformance to material, construction and coverage, type LS2WA only	No failure
Shield - conformance to material, construction and coverage	No failure
Surface transfer impedance	
Milliohms per meter, maximum	700
EMP response time - ms/500 feet	60

Source: https://assist.dla.mil -- Downloaded: 202610-20T00:13Z
Check the source to verify that this is the current version before use.

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

		Requirements
	Aging and compatibility (cable) (125 ± 5°C).....	No failure
(B)	Abrasion resistance (jacket) - scrapes, minimum.....	75
	Acid gas equivalent - percent, maximum	
	Jacket.....	2
	Fillers.....	2
	Insulation.....	18
	Halogen content - percent, maximum	
	Jacket.....	0.2
	Fillers.....	0.2
	Flame propagation (cable).....	No failure
	Immersion (jacket)	
	Tensile strength - percent of unaged, minimum.....	50
	Elongation - percent of unaged, minimum.....	50
	Smoke index, maximum	
	Jacket.....	25
	Fillers.....	--
(B)	Insulation.....	45
	Toxicity index, maximum	
	Jacket.....	5
	Fillers.....	5
	Insulation.....	1.5
	Durometer (jacket) - (type A) hardness, minimum.....	80
	Weathering (jacket).....	No failure

UNIT ORDERING LENGTH: 1000 feet (nominal)

NOTE: Not for Air Force use.

Custodians:
Army - MI
Navy - SH

Preparing activity:
Navy - SH
(Project 6145-1005-46)

Review activities:
Army - AV, CR, ER
Navy - EC
DLA - IS

User activities:
Army - ME, AR, AL
Navy - CG

Source: https://assist.dla.mil -- Downloaded: 202640-20T00:13Z
Check the source to verify that this is the current version before use.